

Generating Highlight Videos of a User-Specified Length using Most Replayed Data

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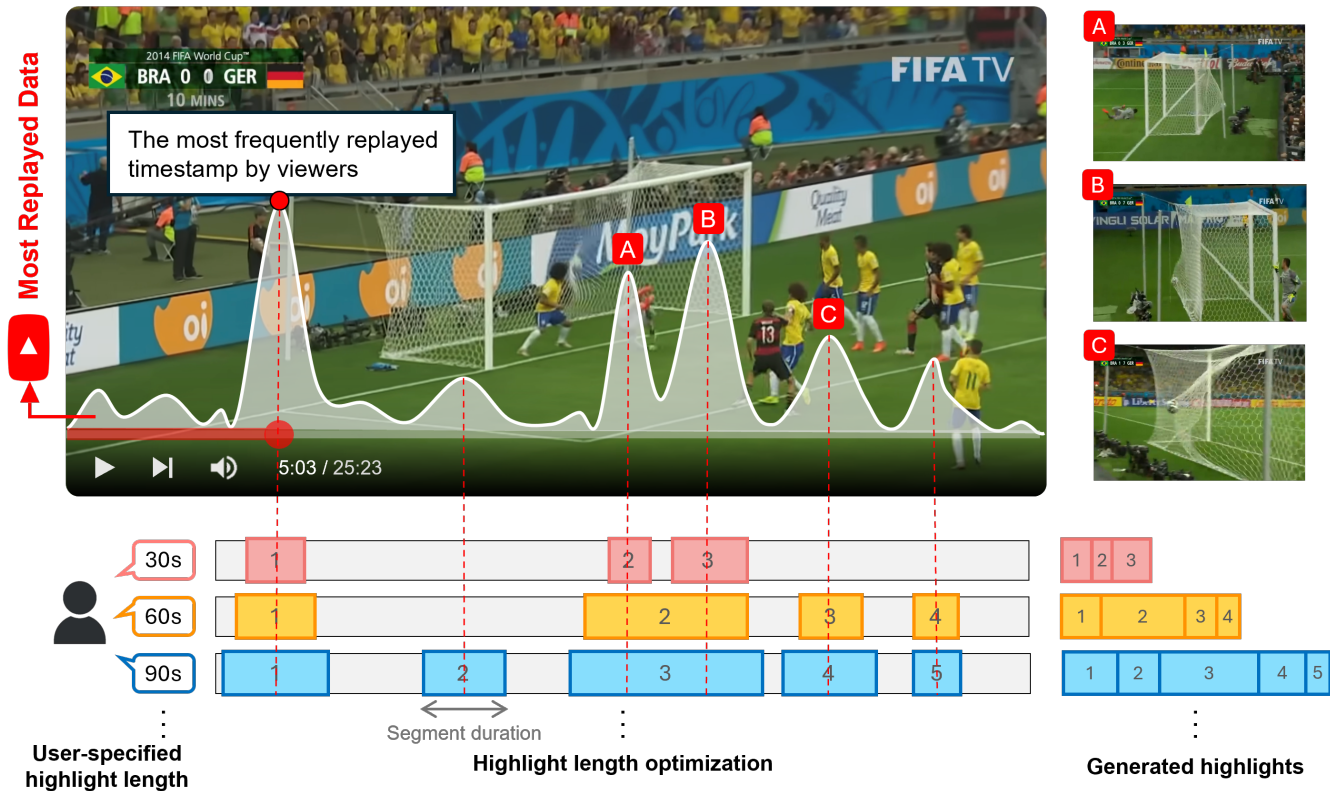


Figure 1: Overview of our method that automatically generates highlights with user-specified lengths using Most Replayed Data (MRD) from YouTube. A, B, and C at the peak of the MRD represent key moments in the soccer match. Image source: [24].

Abstract

A *highlight* is a short edit of the original video that includes the most engaging moments. Given the rigid timing of TV commercial slots and length limits of social media uploads, generating highlights of specific lengths is crucial. Previous research on automatic highlight generation often overlooked the control over the duration of the

final video, producing highlights of arbitrary lengths. We propose a novel system that automatically generates highlights of any user-specified length. Our system leverages *Most Replayed Data* (MRD), which identifies how frequently a video has been watched over time, to gauge the most engaging parts. It then optimizes the final editing path by adjusting internal segment durations. We evaluated the quality of our system's outputs through two user studies, including a comparison with highlights created by human editors. Results show that our system can automatically produce highlights that are indistinguishable from those created by humans in viewing experience.

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CHI '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1394-1/25/04

<https://doi.org/10.1145/3706598.3713880>

CCS Concepts

• Human-centered computing; • Software and its engineering;
• Computing methodologies → Computer graphics;

Keywords

Highlight Generation, Video Summarization, Video Editing, Image Processing, Most Replayed Data

ACM Reference Format:

Minsun Kim, Dawon Lee, and Junyong Noh. 2025. Generating Highlight Videos of a User-Specified Length using Most Replayed Data. In *CHI Conference on Human Factors in Computing Systems (CHI '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 13 pages. <https://doi.org/10.1145/3706598.3713880>

1 Introduction

A *highlight* is a short edit of the original video, which includes the most engaging or important moments. For instance, in sports videos like soccer, highlights generally capture significant events such as shots on target or goals, and in entertainment videos, they feature amusing scenes or notable interactions. Because highlights let viewers capture the key moments in a few seconds or minutes, they are often utilized in TV news as a footage of summarizing a specific event, such as the key plays or score of a game. Highlights are also easily found in many social media platforms because they align well with the content characteristics of these platforms, showcasing intriguing moments in a brief format.

Highlights often need to be created to fit a specific length. In the case of TV broadcasts, they must be produced to fit the exact allotted time slot and adhere to strict advertising or program schedules, making it essential to match the length strictly down to the second. When a video clip is posted to social media, each platform has its own maximum video length limit for uploads, ranging from 60 seconds to 60 minutes¹. Because the preferred duration for content consumption varies across platforms [12, 13, 43], it is important to create highlights that respect these time constraints.

Depending on the final length of the highlight, the number of key moments that will be included can vary. Generally, longer highlights can accommodate more key moments; a 60-second highlight might include 7 key moments compared to 4 in a 30-second highlight. In addition, the duration for each moment can also vary independently depending on the scene content; a 30-second highlight and a 50-second highlight can have the same number of segments with different durations of individual segments. These variations require editors to engage in a time-consuming, iterative process of trial-and-error to adjust the number and duration of moments to fit the desired final length.

While several automatic methods have been proposed for creating highlights or summarized videos, they mostly focus on extracting key scenes from videos, which results in the final output of an arbitrary length [5, 6, 33, 45, 74, 89–91] except for a unified single approach [9, 42, 53] that creates highlights according to the (user)-specified length. Because the approach relies on a simple threshold applied to the assigned importance scores when selecting frames, the resulting highlights may include visually jarring rapid cuts of less than one second, making it difficult for viewers to understand the context of the moment. For example, a soccer match highlight might include segments taken just one second before the ball is

kicked, without providing the context of the preceding dribbling or if the goal was made.

We present a novel method that can automatically generate highlights of a user-specified length with optimized internal segment durations (Figure 1). To decide which frames of the original video to select for a highlight, we leverage *Most Replayed Data* (MRD) available on *YouTube*², which indicates how frequently a video has been watched over time. The final result is obtained by optimizing an objective function that recognizes the most engaging parts via MRD and determining their segment durations properly. The optimization process is performed using dynamic programming [8, 34]. To demonstrate the quality of our method, we conducted comprehensive user studies comparing the highlights created by ours with those created by previous automated methods and by human editors. The results show that our method significantly outperforms existing automated methods and the quality of the results is comparable to that of the results created by human editors in terms of viewing satisfaction.

In the following sections, we explore the implications of MRD as a novel indicator for highlight creation (Section 2). We then introduce related work (Section 3) and describe the method for automatically generating highlights of a user-specified length (Section 4). Next, we showcase our results (Section 5) and evaluate them (Section 6). Finally, we discuss the usefulness and future extensions (Section 7) of our method, including the application of MRD prediction models [19, 73] to original videos with unpublished MRD.

2 Implications of Most Replayed Data as an Indicator for Highlight Creation

In May 2022, *YouTube* introduced a feature called MRD, which visualizes the frequency of viewer replays over time using a line chart on the video player interface (see Figure 1, top). MRD serves multiple purposes for both viewers and content creators. For viewers, it facilitates efficient content consumption by enabling quick navigation to the most engaging parts of a video. For content creators, MRD provides valuable insights into viewer engagement. It reveals which parts of their videos resonate most with viewers, helping creators understand their viewers' preferences and interests. This data can be used to identify less engaging or potentially boring sections, guiding them in refining their future content strategy.

2.1 Reflection of Key Events

In highlight creation, it is essential to include key events or the most engaging scenes of the original video. To this end, we analyzed MRD of a soccer video (2018 World Cup match between France and Argentina [25]) that has clearly marked match events with their time stamps, including goals. Figure 2 shows an MRD graph (black curve) along with the match events (vertical red lines). The figure demonstrates that the timings of match events and MRD peaks, characterized by significant MRD value spikes (i.e., 43, 52, 62, 68, 73, and 97 minutes), align closely. Unlike other MRD peaks

¹ *YouTube Shorts*: 60s; *Naver Clip*: 120s; *X (Twitter)*: 140s; *Threads*: 5m; *Instagram Reels*: 15m; *TikTok*: 60m (accessed on August 23, 2024)

² Generally, *YouTube* provides MRD data for videos with sufficient views, though the exact threshold is not officially specified [28]. Newly released videos may lack this data, as it is generated based on accumulated viewer activity over time.

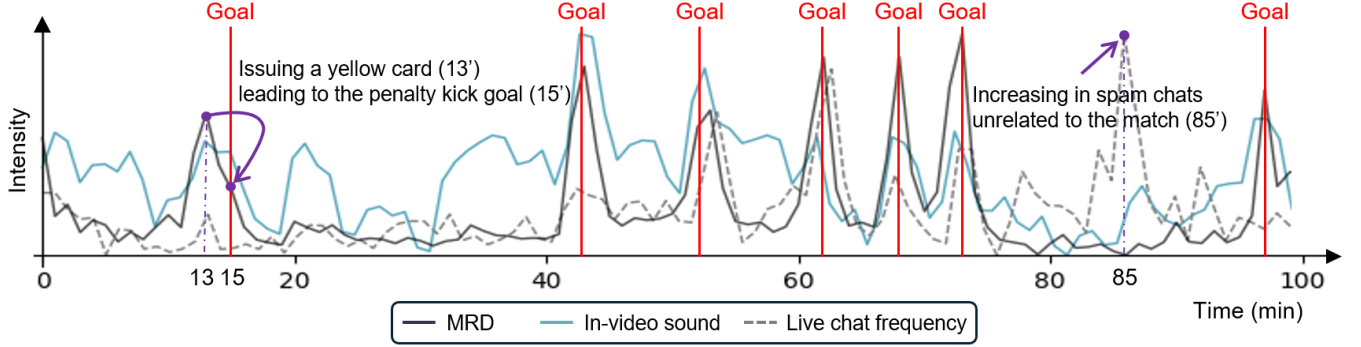


Figure 2: The similar trend of MRD with other known engaging factors and its successful reflection of important events in a soccer match. The in-video sound was calculated using Root Mean Square (RMS) loudness and moving average smoothing was applied. The live chat frequency was derived from crawled *YouTube* live chat data, which counts the number of chats entered per minute and expresses it as a rate of change. All data are normalized to a 0-1 range.

that correspond to a goal, a peak at 13 minutes is associated with a moment of the referee issuing a yellow card, resulting in a penalty kick. This incident proved to be a pivotal turning point in the match, leading to the first goal. This shows that the MRD peaks can be a great indicator for important events.

2.2 Relation to Other Engaging Factors

We also analyzed the correlation between MRD and other known engaging factors, such as in-video sound [5, 60, 79] and live *YouTube* chat frequency [38, 49]. As shown in Figure 2, the shape of the MRD graph is very similar to the those of both the in-video sound (blue curve) and chat frequency (gray dotted curve) graphs. The cosine similarity between the each two graphs is 0.79 (MRD – in-video sound) and 0.70 (MRD – live chat frequency), indicating a high similarity.

Among the three graphs, the MRD graph aligns much better with the events at peak points compared to the other two graphs. For instance, between 20 and 40 minutes, there are no recorded match events, yet the in-video sound graph shows significant fluctuations. Similarly, the chat frequency peaks sharply around 85 minutes due to an increase in chats unrelated to the current match. Users are spamming the chat with off-topic content, such as posting flag emojis of countries not involved in the ongoing match. In contrast, the MRD graph shows distinct peaks primarily at the moments of significant soccer events only for this game, demonstrating a close alignment with the actual flow of the game. We also observed similar results for other video data (refer to the supplementary paper for more details). As a result, we suggest that MRD can be a useful indicator for highlight creation.

3 Related Work

3.1 Automatic Highlight (or Summarized Video) Generation

There are numerous methods for automatically generating highlights (or summarized videos) from original footage [2, 3, 5, 6,

17, 20, 21, 29, 32, 33, 35, 42, 45–48, 63–65, 74, 76, 77, 84, 85, 89–92, 95, 97–99, 101]. These methods aim to create a condensed version of the original video by including important frames or segments. Conventional approaches typically use predefined criteria such as color [97, 98], motion [46, 63], speech [76, 77], or object detection [84] to identify important moments in the original video. In recent years, deep learning approaches have gained prominence, employing neural networks to extract features from visual data [3, 6, 17, 20, 21, 35, 42, 64, 91, 99] or from multimodal data [5, 32, 47, 101] that combines visual information with other modalities such as audio or text, to identify significant moments in the original video.

3.1.1 Automatic Highlight Generation of a User-specified Length.

One of the widely employed approaches to generating highlights of a specified length operates at the segment level [6, 21, 33, 35, 64, 91]. First, importance scores are assigned to each frame based on extracted features. Following this, the video is segmented either using fixed-length intervals or variable-length segments determined by shot boundaries. The highlights are then created by selecting a pre-defined number of top-scored segments. This segment-based approach has limitations because the final length of the highlight varies depending on the length of the selected segments, and due to the segment-based nature, it is challenging to generate highlights conforming to a precise length.

A few methodologies consider individual frames instead of using segments, allowing for more precise control over the highlight length [9, 42, 53]. These methods typically function by selecting frames whose importance scores exceed a certain threshold for inclusion in the highlight. Unfortunately, the results of this approach sometimes adversely affect the viewing experience because they often contain abrupt transitions between scenes, which lasts less than a second, particularly around the peaks of high-scoring frames. These abrupt changes between scenes can make it challenging for viewers to grasp the context. Unlike these approaches, our method generates highlights conforming to a user-specified length while optimizing segment durations and the number of key moments.

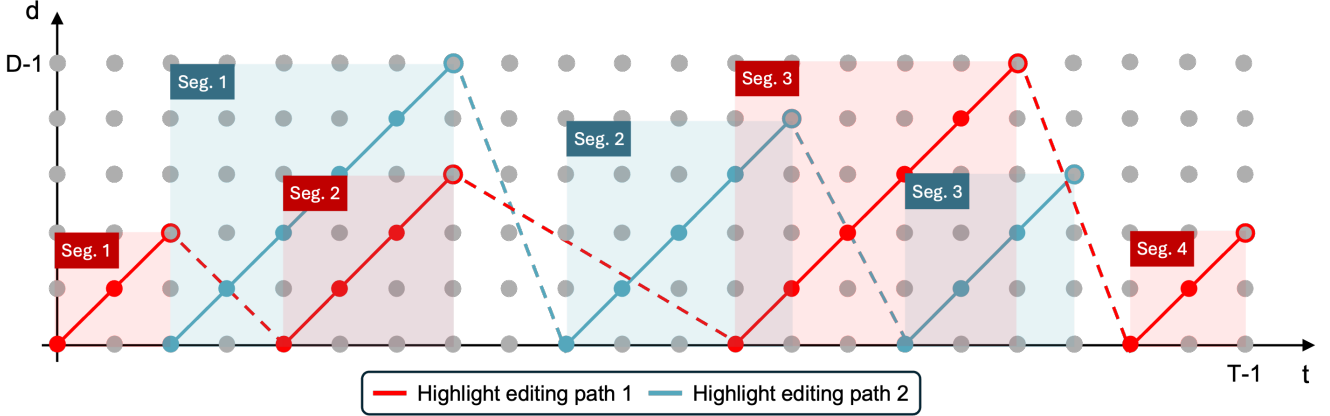


Figure 3: Visualization of the graph construction and highlight editing paths. Seg. refers to segment, the selected highlight region in the original video. We represent each node (gray circles) in two dimensions, $\{t, d\}$, where t denotes the time index of the original video (with T being the total length of the video) and d represents the duration of the current segment that has been played (with D being the maximum segment duration). Two differently colored solid circles and lines represent selected nodes and possible editing path scenarios when L_{user} is set to 12. The editing paths select the next node within a segment according to edge rules, as indicated by solid lines, and transition to the next segment, as indicated by hollow circles and dotted lines. Note that for visual clarity, the rule $t' \geq t + 5$ (when $t' \neq t + 1$) has been relaxed to $t' \geq t + 2$ in this illustration.

3.2 Highlight Generation Reflecting User Engagement

Considering user engagement and their actual interest is crucial when generating highlights [15]. Early studies relied on video content characteristics based on algorithmic decisions, such as notable changes in motion [55] and color [4, 26], as well as event detection [40, 68] and object detection [7]. While these visual cue-based approaches successfully produce highlights that faithfully reflect given objective functions, it is not clear if the resulting highlights align with actual viewer preferences or effectively capture viewers' engagement [30]. Subsequent research utilized viewers' behavioral responses, such as eye gaze and head movement, to better understand and incorporate viewer engagement in highlight generation [39, 62, 88]. Recently, researchers began incorporating actual viewer interest into deep learning approaches, training models on datasets like SumMe [31] and TVSum [69]. These datasets contain videos with importance scores assigned by human raters, closely reflecting real viewer preferences. These approaches employ a small number of annotators (typically 5 to 20) due to the high cost of labeling at the frame or segment level [73]. Therefore, the size of the resulting datasets can be insufficient to reflect diverse user preferences. We utilize MRD, which is based on data from numerous users' real viewing behavior and therefore sufficiently represents the actual viewers' interest.

4 The Method: Automatic-Generation of Highlights given a User-Specified Length

Given the original video and its MRD, our method automatically generates a highlight with a length specified by the user L_{user} . L_{user}

can be any value that is shorter than the length of the original video. We approach the problem of finding the optimal highlight editing path (a sequence of selected frames) by formulating it as a graph structure composed of nodes and edges (Figure 3). Each node represents a timestamp in the original video, and each edge represents a possible transition, indicating how one frame progresses to the next. We assign a reward to the nodes, which favors high MRD values and proper segment duration, and calculate the total reward for all possible paths. We then obtain the optimal highlight editing path that has the highest total reward. In the following, we detail the graph construction (Section 4.1), the design of the reward function (Section 4.2), and the optimization process (Section 4.3). Additionally, our method allows for customization by automatically generating highlights with an option to include or exclude specific frames selected by the user. The customization process is explained in Section 4.4.

4.1 Graph Construction

Figure 3 illustrates our graph construction.

4.1.1 Node. To account for all possible editing paths, we represent node n in two dimensions $\{t, d\}$, where t denotes the time index in seconds of the original video and d indicates the duration of the current segment that has been played (i.e., elapsed time since the last transition) of the highlight. The total number of nodes is $T \times D$, where T is the total number of time points in the original video. D is the maximum value of d , which is set to 30 (equivalent to 30 seconds) to prevent boredom caused by overly long segments for the viewers [16].

4.1.2 Edge. Edge is designed to connect a node $n = \{t, d\}$ to the next node $n' = \{t', d'\}$. The edges are created based on the following rules.

- $t' > t$
: The highlight progresses in chronological order (without rewinding).
- If $t' = t + 1$, then $d' = d + 1$; otherwise ($t' \neq t + 1$), $d' = 0$.
: The condition $t' \neq t + 1$ indicates that the segment has changed. According to the definition of d , if the segment changes, d' is reset to 0.
- If $t' \neq t + 1$, then $t' \geq t + 5$.
: We empirically set a minimum interval of 5 seconds between segments to ensure clear differentiation drawing inspiration from jump cuts in film [66]. This helps users to perceive each segment distinctly, ensuring that different key moments are included in separate segments. Without this rule, users may perceive two consecutive segments as abruptly interrupted with only a one-second interval.

4.1.3 Highlight Editing Path. A highlight editing path consists of a sequence of connected nodes and edges. The total number of nodes and edges in the highlight editing path are both equal to L_{user} .

4.2 Reward Design

The reward R assigned to each node is designed as follows:

$$R(n) = R_{MRD}(t) + wR_{dur}(d). \quad (1)$$

The total reward is a combination of the reward for MRD value at time t in the original video and the reward for the current segment's duration in the highlight. This formulation ensures that frames with high MRD values are included in the highlight, while properly regulating the duration of the segments containing these frames. The weight w is empirically set to 0.2 to balance the influence of MRD values and segment duration reward.

4.2.1 MRD Reward. The MRD graph can be obtained by crawling data from publicly accessible *YouTube* video player interface. The x-axis of MRD graph represents the time. Regardless of the original video's length, the MRD graph provides 100 data points along the time axis. To standardize the data to fit the total number of t , we interpolate these points to a one-second interval. The y-axis indicates the intensity of the MRD values, normalized between 0 and 1. $R_{MRD}(t)$ reflects the y-value of the graph.

4.2.2 Segment Duration Reward. To adjust the duration of segments of the highlight, R_{dur} is applied only to the nodes at which the segment changes ($d' = 0$), as follows:

$$R_{dur}(d) = \begin{cases} R_{ctx}(d) & \text{if } d' = 0 \\ 0 & \text{if } d' \neq 0 \end{cases}.$$

R_{ctx} is set to prevent situations where a segment duration becomes too short or equivalently where too frequent changes may occur [44, 56]. It is defined as follows:

$$R_{ctx}(d) = \begin{cases} 2e^{-\frac{(d-\tau)^2}{8}} - 1 & \text{if } d < \tau \\ 1 & \text{if } d \geq \tau \end{cases}.$$

Here, τ is the recommended minimum duration for a segment, and we set it to 8 (seconds) based on cinematographic principles [18].

This equation represents that the reward increases until d reaches τ , and the value remains at 1 afterwards, thereby promoting segments to last 8 seconds at least.

4.3 Path Optimization

We compute the optimal highlight editing path that maximizes the sum of the rewards via the following equation:

$$P^* = \arg \max_P (R(n_1) + R(n_2) + \dots + R(n_{L_{user}})),$$

where P represents the highlight editing path that consists of a sequence of nodes $\{n_1, n_2, \dots, n_{L_{user}}\}$. We calculate the total sum of rewards for all possible editing paths, having a total number of nodes and edges each equal to L_{user} . The optimal path P^* can be obtained using the dynamic programming method [8, 34].

4.4 User-Driven Highlight Customization

On top of the fully-automated function, we additionally allow *user-driven customization*, enabling users to select the frames they want to include or exclude when they desire further editing, while also ensuring that the final highlight adheres to the user-specified length L_{user} . For example, in a soccer match, a fan of Messi might want to create a highlight focused specifically on Messi's performance. In a variety show, viewers might want to ensure that particularly interesting moments based on their personal preferences are included in the highlight. Our method achieves this goal by deliberately increasing (for inclusion) or decreasing (for exclusion) MRD values at those selected timestamps. Specifically, the reward function $R_{MRD}(t)$ in Equation (1) is modified to $R_{MRD}(t) + R_{user}(t)$. R_{user} is defined as follows:

$$R_{user}(t) = \pm \lambda G(t - \alpha) \quad (+ \text{ for inclusion; } - \text{ for exclusion}). \quad (2)$$

$G(t - \alpha)$ is a Gaussian function centered at the timestamp α . We empirically set the standard deviation of the function as 1.333 and λ as 10. To include or exclude multiple sets of timestamps, multiple $R_{user}(t)$ functions can be added within their corresponding ranges as $R_{MRD}(t) + R_{user_1}(t) + R_{user_2}(t) + \dots + R_{user_n}(t)$.

5 Results

Figure 4 shows the results of highlights generated by our method for various original videos. The method selected key highlight moments across different types of content. For archery, the method effectively captured the moments when the winner scored points and achieved victory. The highlight of the concert featured important instances of large stage props moving, hit songs being performed, and new guests appearing. The highlight of the soccer match successfully extracted the moments of scoring and near-miss saves by the goalies. The highlight of the audition illustrated Taekwondo board-breaking scenes and moments when contestants received the highest possible praise, signified by a special buzzer sound indicating exceptional performance. For more video results, please refer to our project page: <https://w-dlee.github.io/highlights>.

5.1 Results on a User-Specified Length

Figure 5 shows how the segments differ when highlights are generated using four different L_{user} for the same original video. When

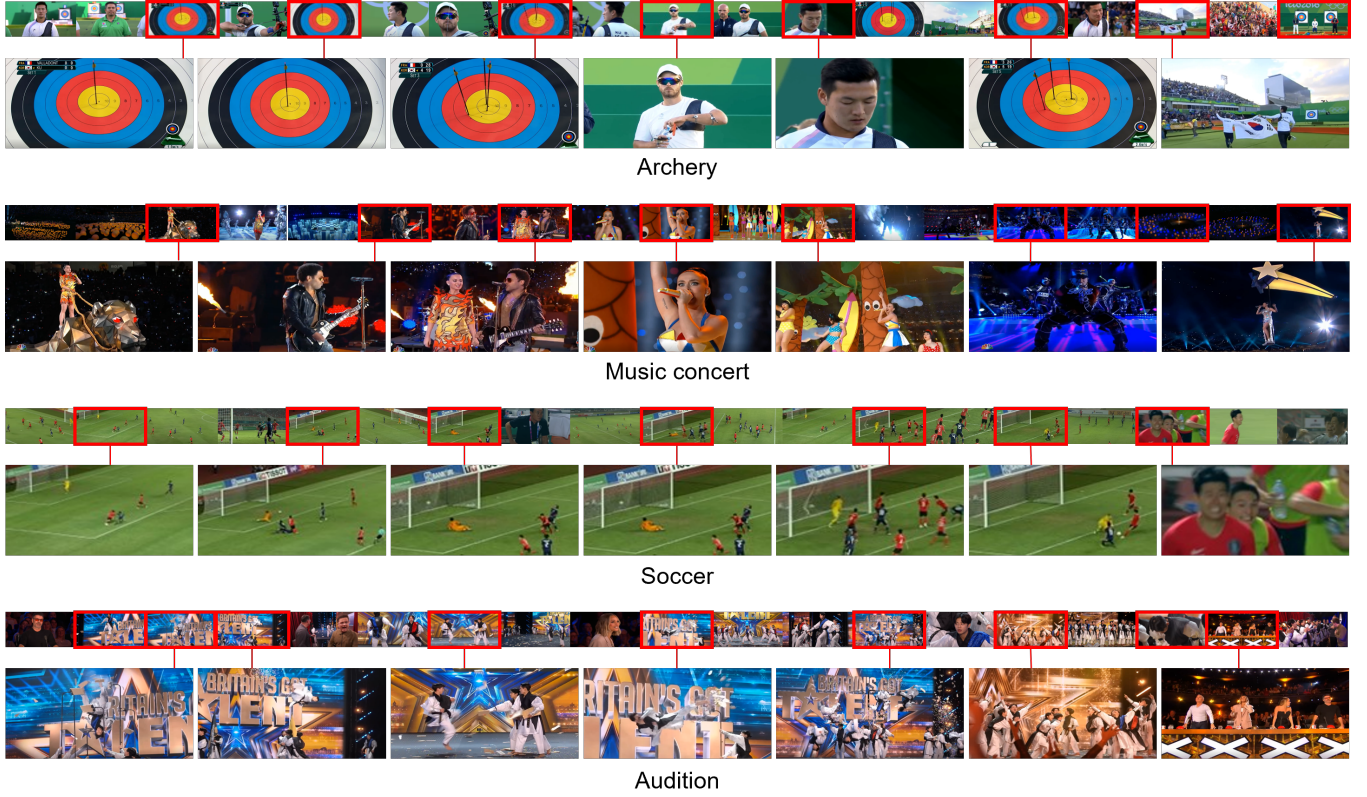


Figure 4: Example visual results of highlights generated by our method. The top row of each example shows the original video frames from archery, music concert, soccer match, and audition. The bottom row displays extracted key moment frames when highlights were generated with L_{user} being 120. Image source: archery [59], music concert [57], soccer match [80], and audition [75].

L_{user} is 30, the method selected segments with the top three highest MRD peaks. As L_{user} increased, the method not only extended the durations of the top three selected segments but also included additional segments. To further understand how segments change according to L_{user} , we analyzed the average segment count, segment duration, and MRD value (see Figure 6). Five YouTube videos were used as the original videos, including the one used for Figure 5. As expected, the average segment count and segment duration increased with the increase of L_{user} . The average segment duration was over 8 seconds for all L_{user} values due to the enforcement by the duration reward described in Section 4.2.2. The average MRD value decreased as L_{user} increased. This occurred because the method initially selected segments with the highest MRD peaks and then gradually incorporated the region with lower MRD values around these peaks as L_{user} increased, resulting in a decrease in the overall average MRD value. These results indicate that our method successfully performs optimization for the given L_{user} by selecting the appropriate number of segments and their durations.

5.2 Results on User-Driven Customization

Figure 7 shows the results of our user-driven highlight customization. As can be seen in the figure, before the customization, the

resulting highlight included segments with the highest MRD peaks. After the user selected their preferred timestamps (4m 24s, 7m 11s, 9m 19s, 10m 42s, and 11m 57s), the highlight was changed to include the segments containing them. For the remaining duration, segments with the highest MRD peaks (1m 46s, 3m 33s, and 6m 3s) were partially included, while keeping the total length of the output consistent with L_{user} .

5.3 Computation Time

Table 1 shows the computation time of our method. The computation time increased with the length of the original video and L_{user} . The longer the original video, the more the nodes were created, and similarly the longer L_{user} was, the more the edges were created. Both of these factors increased the number of possible editing paths that need to be computed. For example, it required approximately 5 seconds to generate a 1-minute highlight from a 2-minute original video. On the other hand, it required approximately 10 minutes to generate a 3-minute highlight from a 10-minute original video. The computation time was measured on a computer with a 2.20 GHz Intel Xeon Silver 4214 processor and 32 GB of memory. The method was implemented in Python.

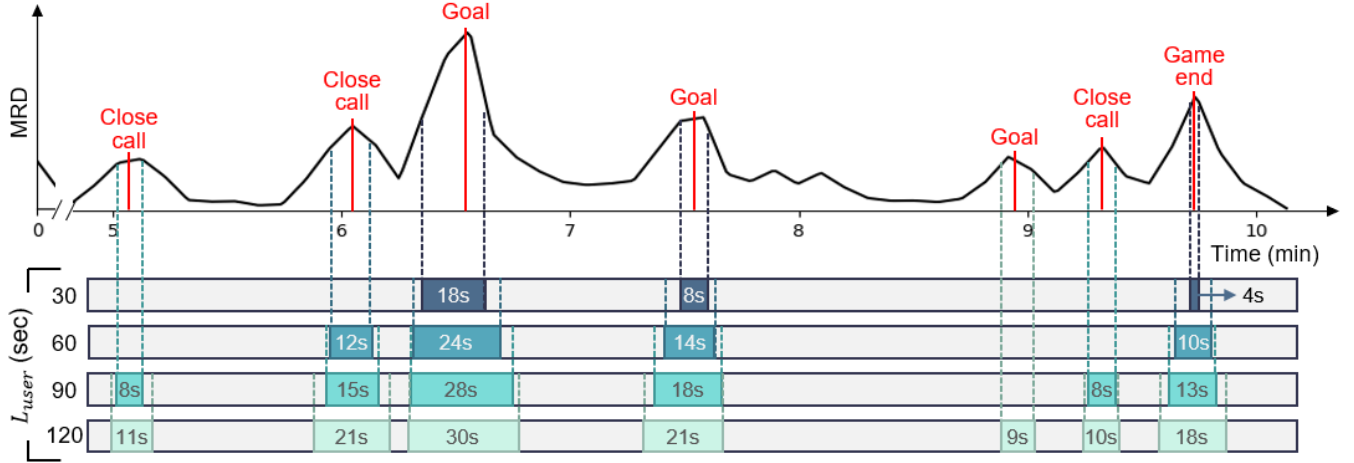


Figure 5: Highlight segment selection results created by using different L_{user} (30, 60, 90, and 120). The graph illustrates the change of MRD values over a soccer match [80], with key soccer moments marked.

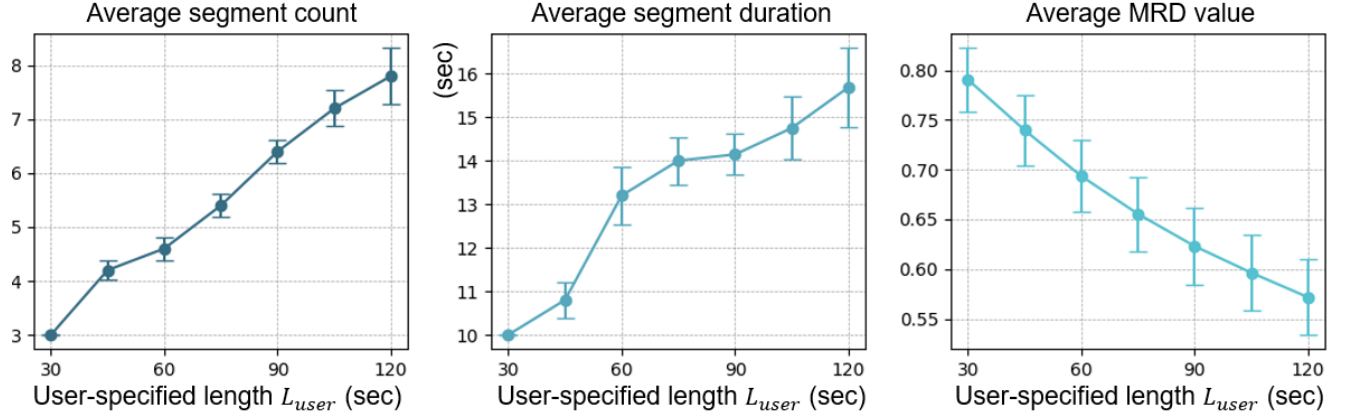


Figure 6: The graphs of average segment count, average segment duration, and average MRD changes corresponding to the changes of L_{user} . Each data point represents the average results from five YouTube videos [54, 58, 78, 80, 87] and error bars indicate the standard deviation of the mean.

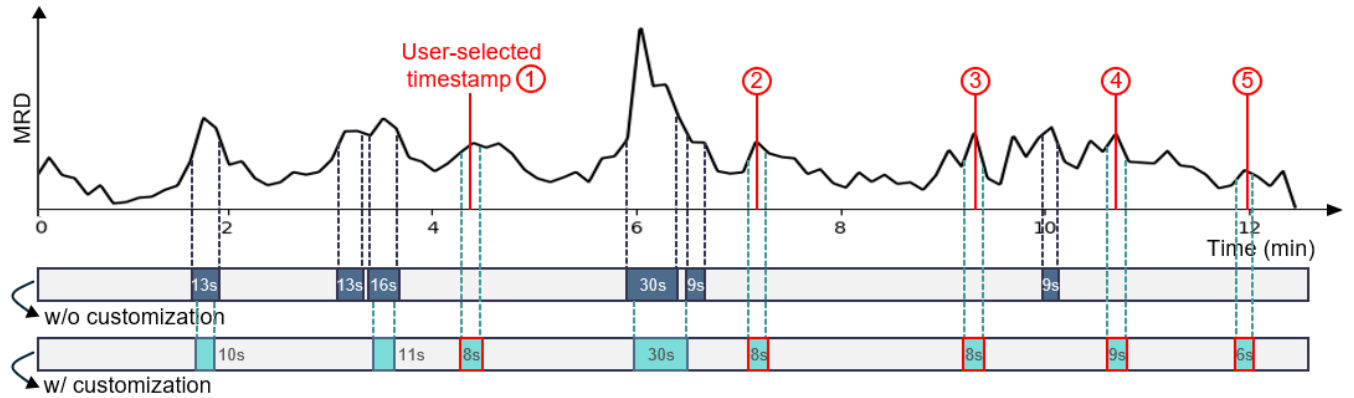


Figure 7: The result of user-driven highlight customization. We used a music concert video [57] and L_{user} being 90 as input. Without customization, the highlight primarily included segments with high MRD values. When the user selected specific timestamps, our method generated a highlight that included those specific segments.

Table 1: Computation time for highlight generation with various original video lengths and user-specified highlight lengths (L_{user}). The computation time includes only the path optimization process, excluding video rendering time.

Video	Original video length	L_{user}	Computation time
[22]	2m 10s	30s	2.72s
		1m	4.34s
[11]	3m 15s	30s	6.49s
		1m	15.9s
[23]	5m 5s	1m	41.62s
		1m 30s	59.7s
[75]	8m 40s	1m	2m 32s
		1m 30s	3m 51s
		2m	4m 58s
[71]	10m 21s	1m	3m 24s
		2m	7m 16s
		3m	9m 59s

6 Evaluation

We conducted two comprehensive user studies to demonstrate the effectiveness of our method in generating high-quality highlights. In the first study, our objective is to evaluate methods for generating highlights of a user-specified length when given the same importance scores. Therefore, we compared our length optimization method with that of a previous method, using the same MRD as input (Section 6.1). In the second study, we compared highlights generated by our method with those created by humans to assess how comparable the quality of our automatically generated highlights is compared to that of human-edited ones (Section 6.2). A total of 30 participants (15 males, 15 females; age: $M = 23.40$, $SD = 3.37$) participated in both Study 1 and Study 2. Each study required approximately 25 minutes, for a total duration of 50 minutes, and was conducted using an online form. The participants were compensated \$7.5 for their time.

6.1 User Study 1: Our Method vs. Previous Methods

6.1.1 Setting. We collected five YouTube videos [27, 70, 72, 83, 86] and generated three types of highlights: our method, a threshold-based method [9, 42, 53], and a random method. We identically set L_{user} to 60 for all the videos and methods. The threshold-based method, which is a unified single approach used for generating highlights in a user-specified length, arranged the frames in a descending order of MRD and selected frames from high MRD to low MRD until L_{user} was reached, and then restored those selected frames to their original order to generate the highlights. The random method randomly selected k non-overlapping segments from the original video, where k equals the number of segments in highlights generated by our method. The duration s of each segment was determined to satisfy the equation $k \cdot s = L_{user}$.

Participants watched all three versions of highlights for each of the five original videos³. They were instructed to watch each of the highlights with sound on in its entirety without skipping. The presentation order of the five sets of highlights was randomized, and for each set, the presentation order of the three versions of highlights was also randomized to minimize the bias from the ordering. After watching each highlight, the participants were asked to respond to the following three questions using a 7-point Likert scale to rate its appropriateness, key moment inclusion, and satisfaction. The questions were as follows, with descriptors provided for points 1, 4, and 7.

- Q1. **Appropriateness:** Do you think the duration of each segment is appropriate?
(1: Very inappropriate; 4: Neutral; 7: Very appropriate)
- Q2. **Key moment inclusion:** Do the segments include sufficiently interesting and important moments, regardless of their duration?
(1: Not at all included; 4: Neutral; 7: Very well included)
- Q3. **Satisfaction:** How satisfied are you with the overall viewing experience?
(1: Very dissatisfied; 4: Neutral; 7: Very satisfied)

6.1.2 Results. Figure 8 shows the results. For each question, we conducted the Friedman tests, and the results revealed significant differences ($p < 0.05$) among the three methods across all questions (Q1: $\chi^2(2) = 37.16$; Q2: $\chi^2(2) = 40.99$; Q3: $\chi^2(2) = 38.97$; $p < 0.0001$ for all). We performed post-hoc analyses using the two-tailed Wilcoxon signed-rank tests with Holm-Bonferroni correction.

For appropriateness (Q1), our method ($M = 5.27$) significantly outperformed both the threshold ($M = 4.14$; $p < 0.001$) and random methods ($M = 4.06$; $p < 0.001$). The threshold method's lower performance can be attributed to its naive approach of selecting high MRD values without optimizing segment length, often resulting in jarring segment transitions of one or two seconds (see Figure 9). The random method's poor performance can be explained by the unnatural transitions that occur when segment boundaries in the highlight coincide with shot boundaries in the original video, creating abrupt and disjointed visual shifts.

In terms of key moment inclusion (Q2), our method ($M = 5.59$) obtained a significantly higher rating over both the threshold method ($M = 5.02$; $p < 0.001$) and the random method ($M = 4.15$; $p < 0.001$). Both our method and the threshold method significantly outperformed the random method, demonstrating the effectiveness of MRD-based strategies in capturing important or interesting moments. The difference between our method and the threshold approach can be attributed to the threshold method's tendency to create one or two-second long short segments that do not provide sufficient time for viewers to understand highlights adequately. Regarding satisfaction (Q3), our method ($M = 5.43$) significantly outperformed both the threshold ($M = 4.59$; $p < 0.001$) and random methods ($M = 3.99$; $p < 0.001$). Overall, our method consistently outperformed the baseline approaches and the prior work for all three questions.

³Participants watched only the highlights without viewing the original videos. This approach mirrors real-world scenarios where users seek to grasp interesting content quickly without investing time in full-length videos, and aligns with the experimental designs employed in previous highlight generation research [50, 61].

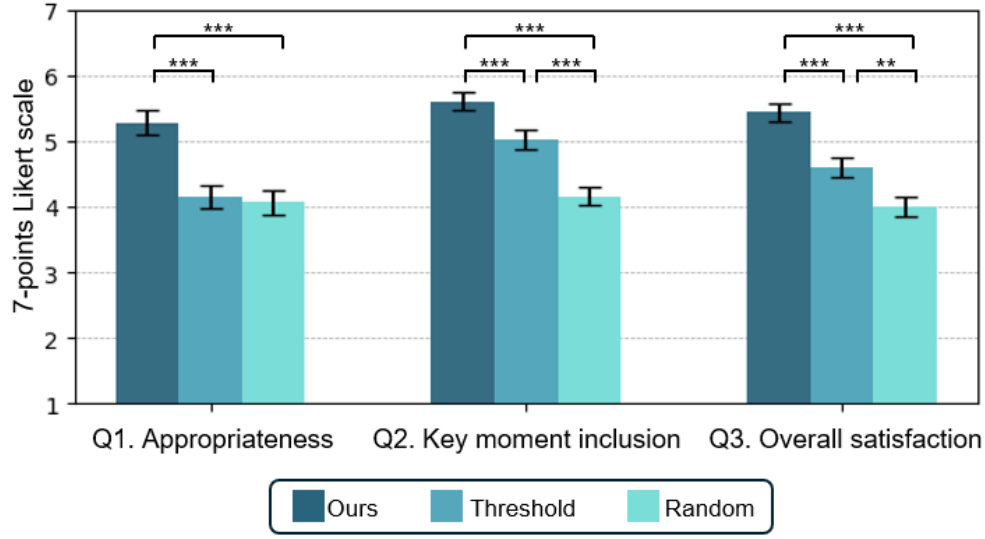


Figure 8: 7-point Likert scale results of three methods for appropriateness (Q1), key moment inclusion (Q2), and overall satisfaction (Q3). Error bars represent the standard error of the mean. Horizontal brackets and asterisks above show statistically significant differences between pairs of methods (*: $p < 0.01$; ***: $p < 0.001$).

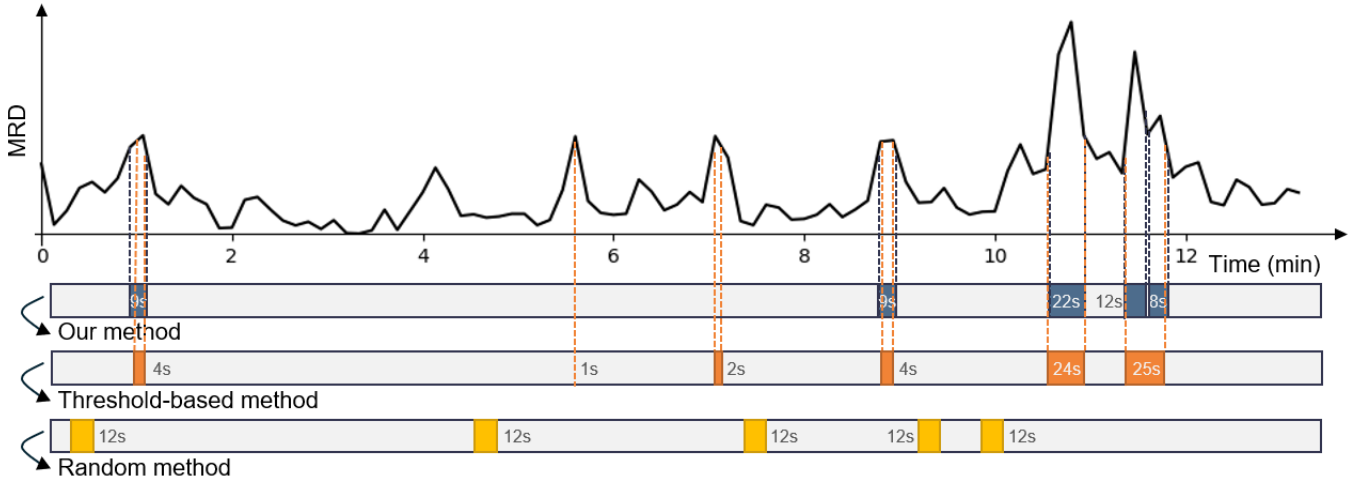


Figure 9: Segment selection results of our method, the threshold-based method, and the random method. This video is from one of the five original videos used in the first user study [83].

6.2 User Study 2: Our Method vs. Human-Editors

6.2.1 Setting. We selected ten pairs of videos, each consisting of a *YouTube* video and its corresponding *YouTube Shorts*. Two versions of highlights were used for comparison: one generated by our method and the other created by human editors (*YouTube Shorts*). For our method-generated highlights, we set L_{user} (ranging from 20 to 60 seconds) to match the length of the corresponding human-edited highlight. For human-edited highlights that had been cropped to fit vertical short-form video formats, we restored them to their original aspect ratio by matching frames with the original

videos. For more details about the selected videos, please refer to the supplementary paper.

Participants watched the two versions of highlights for each of the ten original videos⁴. They were instructed to watch each of the highlights with sound on in its entirety without skipping. The presentation order of the ten sets of highlights was randomized, and for each set, the presentation order of two versions of highlights

⁴Participants watched only the highlights without viewing the original videos. This approach mirrors real-world scenarios where users seek to grasp interesting content quickly without investing time in full-length videos, and aligns with the experimental designs employed in previous highlight generation research [50, 61].

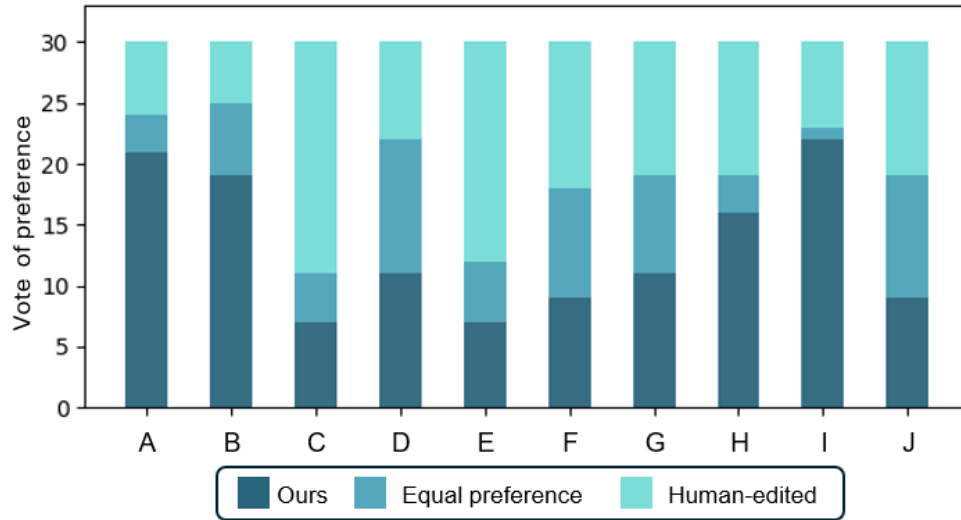


Figure 10: Vote distribution among three options for ten sets of highlight pairs.

was also randomized to minimize the bias from the ordering. After watching each pair, they were asked to vote on their preference: our method-generated highlight, the human-edited highlight, or equal preference. If participants preferred one highlight over the other, they were given an open-ended question to provide reasons for their preference.

6.2.2 Results. Figure 10 shows the vote distribution for each option. We conducted the Friedman test, and the result revealed no statistically significant differences among the three options ($\chi^2(2) = 5.8919, p = 0.0526$). This indicates that there is no strong preference for any particular option among the participants, suggesting that our method produces highlights with quality comparable to the quality of those created by human editors.

The majority of participants noted that the highlights generated by our method effectively captured important moments suitable for a highlight. Specifically, twelve participants (P1, P3, P5-6, P8-9, P11, P19, P22-23, P27, P29) reported that the highlights maintained a clear narrative structure, making it easy to understand the overall flow and storyline. Ten participants (P1, P9, P11, P14-15, P17, P19, P23-24, P29) found our highlights engaging, easily understandable, and interesting. Six participants (P6, P7, P13, P15, P23, P29) commented on the appropriate duration of each segment, and three participants (P6, P12, P27) praised the smooth and natural editing. These diverse positive evaluations demonstrate the effectiveness of our highlight generation method.

7 Limitations and Future Work

7.1 Applications of MRD with Alternative Data

Our method relies on MRD graphs obtained from *YouTube*. However, MRD graphs are currently accessible only for uploaded videos with a sufficient number of views⁵; thus, our method cannot generate highlights for videos with relatively fewer views or videos

that have not been uploaded to *YouTube*. One way to overcome this limitation is to utilize recently developed deep-learning based models [19, 73]. These models can predict MRD graphs from any input video, regardless of view count. To demonstrate feasibility, we present the quality of the predicted MRD in the supplementary paper and show the resulting highlight created by our method using the predicted MRD in the supplementary video⁶. By integrating such models, we expect that our method can generate highlights for any video, even those not uploaded to *YouTube*. While we used MRD graphs to reflect users' actual engaging moments, our method can operate with any type of graph that has annotated intensity values. Therefore, it is possible to generate highlights by replacing MRD graphs with appropriate data graphs for specific situations or purposes; this only requires modifying the graph for R_{MRD} in Equation (1) to the new one. For example, unpublished real-time viewing data graphs from the TV broadcast system or interaction data for educational videos [41] could be utilized.

7.2 Potential Enhancements of the Method

7.2.1 Reducing Computation Time for Overly Long Original Videos. When extremely long original videos (e.g., marathons > 3 hours; golf tournaments > 4 hours; cricket games > 6 hours; live concerts > 8 hours) are input to our method, the computation time can become excessively long, making it impractical to obtain results. This is because the number of nodes proportionally increases according to the length of the original video, leading to an exponential growth in the number of possible editing paths that need to be computed. One way to alleviate this situation would be to adjust the node creation interval, which is currently one-second. Making this interval more sparse (e.g., 10 seconds or 1 minute) would significantly reduce the computation time at the cost of increasing the granularity of the highlight generation, as the method would then work with larger

⁵While *YouTube* provides MRD data for videos with a sufficient number of accumulated views, it does not officially disclose the exact threshold [28].

⁶<https://w-dlee.github.io/highlights>

time units instead of the current one-second precision. Additionally, a promising future direction could be to develop methods for pruning relatively unimportant edges early in the process, instead of performing an exhaustive search [36, 81, 100].

7.2.2 Considering the Original Video’s Shot Boundaries. Because our method does not consider shot boundaries within the original video, the segments of the generated highlights from cut-edited original videos may occasionally include shot boundaries. While the continuous nature of MRD may implicitly cover shot boundaries, for more precise and explicit control over segments, future work could explore leveraging shot boundary detection techniques [10, 96]. For example, removing the edges just before or after detected shot boundaries, at the moment when the segment changes ($d' = 0$), can help ensure smoother transitions and reduce visual discomfort [44].

7.3 Extensions for Advanced Highlight Generation

7.3.1 Highlights in Various Aspect Ratios. Our method generates highlights with the same aspect ratio as the original video. While original videos often have a widescreen aspect ratio, short-form platforms frequently feature highlights with a portrait aspect ratio [82]. Given this characteristic, future developments could include a strategy for automatically creating highlights from the original footage, tailored to specific lengths and specific aspect ratios. To achieve this, methods for automatic aspect ratio conversion based on region-of-interest (ROI) [37] could be used.

7.3.2 Subject-specific Highlights. Subject-specific highlights, featuring intriguing moments of particular individuals within the original video, are gaining popularity [52, 67, 101]. Although our method’s customization feature allows users to select timestamps with specific individuals to generate such highlights, creating highlights for multiple individuals within a single video can still require manual effort. Utilizing person identity detection algorithms [1, 94] could automate this process. By applying outputs from these algorithms to α in Equation (2), we anticipate that subject-specific highlights of a user-specified length can be created.

7.3.3 Narrative-based Highlights. Highlights typically focus on showcasing interesting or engaging moments of the original video instead of fully capturing the original video’s narrative arc or plot structure [73]. This tendency becomes more pronounced, especially when the highlight is of a short user-specified length (e.g., less than 30 seconds) and includes only a small number of segments. When the creation of highlights that faithfully reflect the narrative is desired, the use of the video description models [14, 51, 93] can be beneficial. These models possess the ability to comprehend not only the overall plot of the video but also the specific events occurring at particular timestamps. By integrating descriptions generated by these models into our method, high rewards could be assigned to highly important timestamps, similar to how our highlight customization function operates.

7.4 Unsuitable Video Types for Highlight Creation

Original videos without significant events, and therefore lacking clear peaks in MRD, may result in unsatisfying highlights. This occurs because the highlight segments selected by our method rely on high MRD peaks; when these peaks are scarce, the highlights include few engaging segments, making it difficult to captivate viewers. Examples include videos consisting predominantly of static or minimally changing images displayed for extended periods, content featuring repetitive loops over long durations, or videos unsuitable for highlight creation, such as ambient videos (e.g., long-form nature footage).

8 Conclusion

This paper introduces a novel method for automatically generating highlights of a user-specified length by leveraging MRD. We first demonstrated the suitability of using MRD for highlight creation by investigating its inclusion of key moments and its correlation with other known engaging factors. Building on MRD, our method generates highlights by optimizing an objective function that selects the most engaging parts and determines their durations properly. The optimization process is performed using dynamic programming. To evaluate our method, we conducted two user studies. The first study demonstrated superior viewer satisfaction and engagement with our method compared to previous methods, and the second study showed that our method produces highlights comparable to those created by human editors.

Acknowledgments

This research was supported by Basic Science Research Program through National Research Foundation of Korea (RS-2024-00452748) and by Institute of Information & Communications Technology Planning & Evaluation grant (RS-2024-00439499). We thank the anonymous reviewers for their constructive feedback.

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